8.2 SLAAC

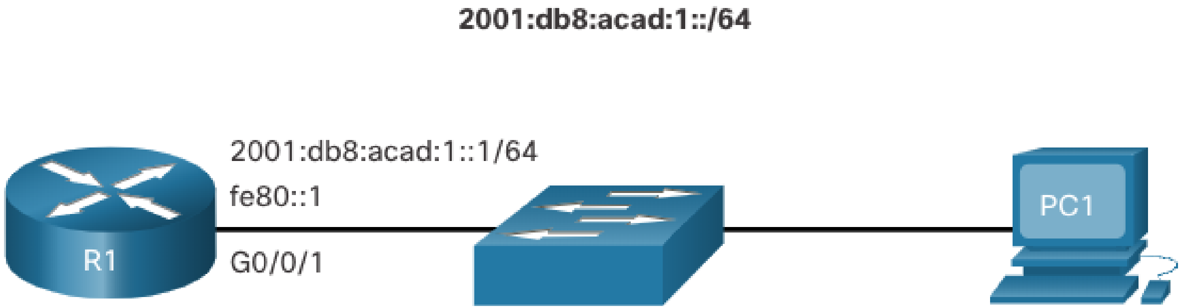
SLAAC Overview

Not every network has access to a DHCPv6 server but every device in an IPv6 network needs a GUA. The SLAAC method enables hosts to create their own unique IPv6 global unicast address without the services of a DHCPv6 server.

- SLAAC is a stateless service which means there is no server that maintains network address information to know which IPv6 addresses are being used and which ones are available.
- SLAAC sends periodic ICMPv6 RA messages (i.e., every 200 seconds) providing addressing and other configuration information for hosts to autoconfigure their IPv6 address based on the information in the RA.
- A host can also send a Router Solicitation (RS) message requesting an RA.
- SLAAC can be deployed as SLAAC only, or SLAAC with DHCPv6.

SLAAC

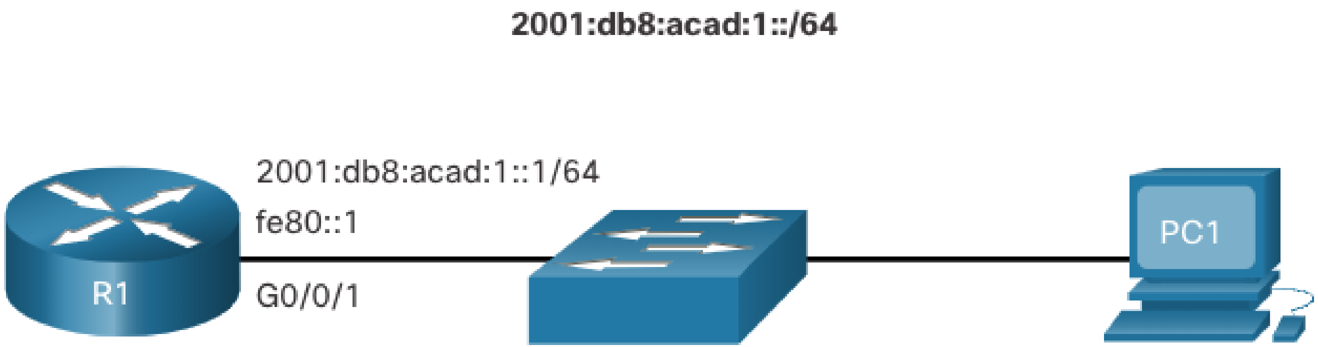
Enabling SLAAC




```
R1# show ipv6 interface G0/0/1
GigabitEthernet0/0/1 is up, line protocol is up
  IPv6 is enabled, link-local address is FE80::1
  No Virtual link-local address(es):
  Description: Link to LAN
  Global unicast address(es):
    2001:DB8:ACAD:1::1, subnet is 2001:DB8:ACAD:1::/64
  Joined group address(es):
    FF02::1
    FF02::1:FF00:1
(output omitted)
R1#
```

```
R1(config)# ipv6 unicast-routing
R1(config)# exit
R1#
```

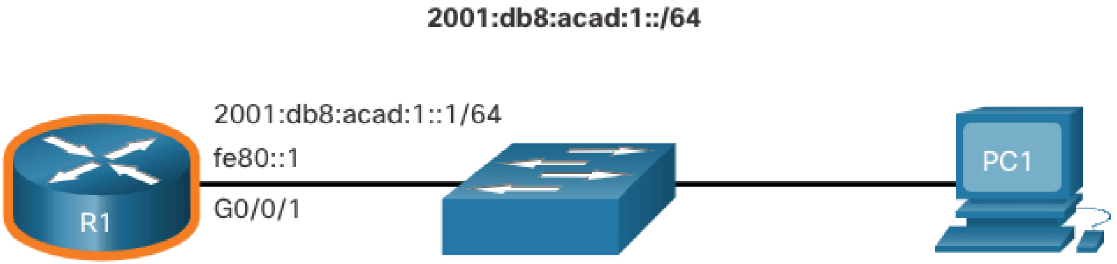
Enabling SLAAC (Cont.)



- R1 will now begin to send RA messages every 200 seconds to the IPv6 all-nodes multicast address ff02::1.

```
R1# show ipv6 interface G0/0/1 | section Joined
  Joined group address(es):
    FF02::1
    FF02::2
    FF02::1:FF00:1
R1#
```


SLAAC Only Method



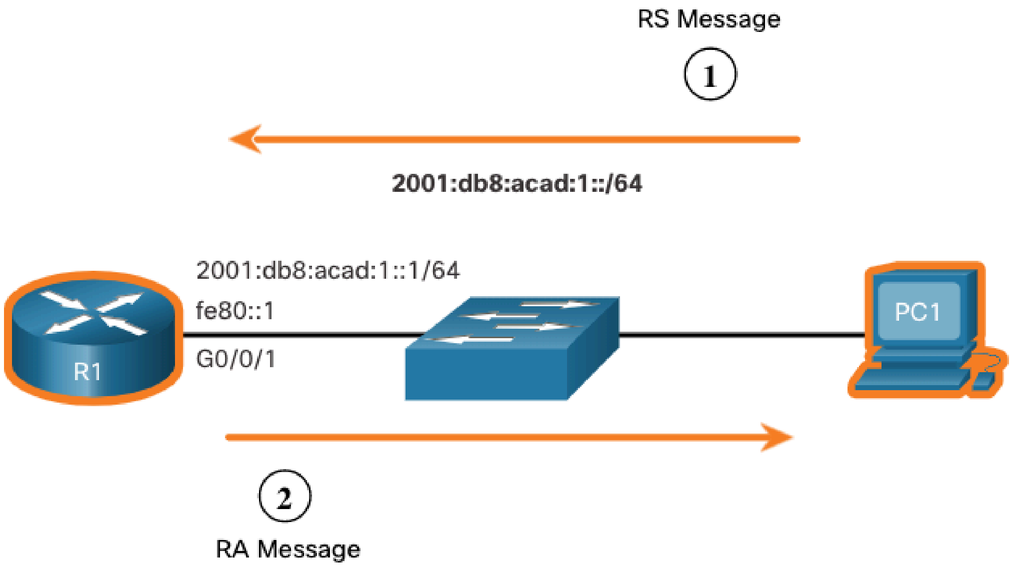
RA Message

Flag	value
A	1
O	0
M	0

```
C:\PC1> ipconfig
Windows IP Configuration
Ethernet adapter Ethernet0:
   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2001:db8:acad:1:1de9:c69:73ee:ca8c
   Link-local IPv6 Address . . . . . : fe80::fb:1d54:839f:f595%21
   IPv4 Address. . . . . . . . . . . : 169.254.202.140
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . . : fe80::1%6
C:\PC1>
```

ICMPv6 RS Messages

1. PC1 has just booted and sends an RS message to the IPv6 all-routers multicast address of ff02::2 requesting an RA.
2. R1 generates an RA and then sends the RA message to the IPv6 all-nodes multicast address of ff02::1. PC1 uses this information to create a unique IPv6 GUA.



Host Process to Generate Interface ID

Using SLAAC, a host acquires its 64-bit IPv6 subnet information from the router RA and must generate the remainder 64-bit interface identifier (ID) using either:

- **Randomly generated**
- **EUI-64** - interface ID using its 48-bit MAC address with insertion of fffe in the middle of the address.

```
C:\PC1> ipconfig
Windows IP Configuration
Ethernet adapter Ethernet0:
   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2001:db8:acad:1:1de9:c69:73ee:ca8c
   Link-local IPv6 Address . . . . . : fe80::fb:1d54:839f:f595%21
   IPv4 Address. . . . . . . . . . . : 169.254.202.140
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . . : fe80::1%6
C:\PC1>
```

Duplicate Address Detection

A SLAAC host may use the following Duplicate Address Detection (DAD) process to ensure that the IPv6 GUA is unique.

- The host sends an ICMPv6 Neighbor Solicitation (NS) message with a specially constructed solicited-node multicast address containing the last 24 bits of IPv6 address of the host.
- If no other devices respond with a Neighbor Advertisement (NA) message, then the address is virtually guaranteed to be unique and can be used by the host.
- If an NA is received by the host, then the address is not unique, and the host must generate a new interface ID to use.